Ghost Data Audit: The Whole Project, Explained

A plain-language guide for explaining the project to anyone. Every technical term is defined in the glossary at the end (section 11).

1. The problem in one paragraph

A fictional shop, **Northstar**, deletes customers' data when they ask. Its dashboard says "deletion COMPLETE", but copies of the data survive in other places: database replicas, search indexes, support tickets, exports, backups. Some background jobs (syncs, backup restores) can even bring deleted data back. These surviving copies are **ghost data**. Our job was to build an **audit agent**. For each deletion request, it checks the evidence and decides whether the deletion is really done. If it isn't, it lists what is left, how to fix it, and how to check the fix.

> **Say it in one line:** "We built a system that checks whether a 'deleted' customer is really gone, using evidence rather than the dashboard's word, and writes a step-by-step fix plan when they aren't."

2. What we were given (the data)

File	What it is	Plain meaning
`deletion_requests.csv`	518 cases to answer	Each row: a person's email, what they asked for, what the dashboard claimed, and the **audit time** (the moment we judge).
`sample_deletion_requests.csv` + `sample_output.csv`	32 cases *with the correct answers*	Our only answer key. We used it to measure and to learn the organizers' conventions.
`user_identities.csv`	Current identity of each user	user_id, name, email, phone, email hash.
`user_audit_log.csv`	Identity history	Old emails and phones; a person may have changed email, so we must search for the old one too.
`system_activity.csv`	Event stream	"Deleted X: SUCCESS", "sync ran", "backup restored". These are **claims**, not proof.
`deletion_rules.csv`	15 policy rules	What to do with each kind of data: delete it, redact it, anonymize it, or keep it.
`evidence_files.csv` + `evidence/`	94 evidence files	SQL database snapshots, JSON exports, logs, PDFs, screenshots. This is where the **truth** is.

The answer format (output.csv)

One row per case: `case_id, deletion_status, steps_json`.

- **COMPLETE**: really deleted. Steps are only *verification* checks, each citing the evidence that proves it.
- **INCOMPLETE**: data remains. Steps are *findings* (what's left, citing evidence), then *remediation* (the fix), then *verification* (checks to run after the fix).
- **NOT_APPLICABLE**: the person never asked for deletion (e.g. "please correct my email"). Steps are `[]`.

3. A real example, start to finish (sample case_16)

```
Request:   diya.k4436@outlook.com, "Please treat this message as a request for permanent
account deletion..."
Dashboard: NOT_STARTED, reported 2026-09-01
Audit at:  2026-09-21 10:00 (we may only use evidence from this moment or earlier)

Activity log says:
  activity_3571  DELETE account_994   -> FAILED
  activity_3580  DELETE search_1007   -> SUCCESS   (a claim!)

Evidence evidence_05 = database snapshot taken 2026-09-21 10:00:
  customer_data_read_replica still contains account_994 (Diya Kapoor, her email)
  customer_data_read_replica still contains search_1007 (her search entry)
  -> the "SUCCESS" delete of search_1007 is a GHOST: the copy survived in a replica.

Correct answer: INCOMPLETE
  FINDING       REPORT_REMAINING_RECORD  account_994  evidence_05
  FINDING       REPORT_REMAINING_RECORD  search_1007  evidence_05
  REMEDIATION   DELETE_RECORD            account_994
  REMEDIATION   DELETE_RECORD            search_1007
  VERIFICATION  VERIFY_RECORD_ABSENT     account_994
  VERIFICATION  VERIFY_RECORD_ABSENT     search_1007
  VERIFICATION  VERIFY_RECORD_PRESENT    audit_record_9528   (security log must be KEPT)
```

Our system produces exactly this answer. It shows the three ideas behind the whole project: **don't trust claims**, **check every copy**, and **keep what the policy requires** (the security log is kept, not deleted).

NOT_APPLICABLE example: a Spanish request, "Corrige mi correo, pero no borres mi cuenta" ("Fix my email, but don't delete my account"), gets `NOT_APPLICABLE, []` even if the dashboard shows a deletion.

4. How our system works (the pipeline)

```
evidence files ──► [1] READ everything ──► "facts" (who/what/where/state/when, plus which
file proves it)
                   [2] INTENT    did the person ask for deletion?   no → NOT_APPLICABLE
                   [3] IDENTITY  current + old emails/phones → the person's user_id
                   [4] COLLECT   every record linked to that person
                   [5] TIME CUT  keep only facts known at or before the audit time
                   [6] POLICY    look up the rule: delete / redact / anonymize / keep / block
/ disable
                   [7] PLAN      findings → fixes (stop syncs/restores first) → checks
                   [8] DOUBLE-CHECK  simulator + validator must both agree
                   ──► output.csv  (only written if every row passes)
```

Step	What happens	Why it matters
1 Read (ingest)	Opens SQL, JSON, logs, PDFs; reads screenshots with OCR. Every observation becomes a **fact** tagged with its source file.	Every answer can be traced to a file: no made-up evidence.
2 Intent	Reads the request text (any language).	Correction or export requests must not trigger deletion.
3 Identity	Finds the person's current and historical identifiers.	Old systems may still store the old email.
5 Time cut	Ignores anything captured after the audit time.	No "future knowledge"; the judgement is fair at that moment.
6 Policy	Applies the 15 rules (e.g. a support message shared with an agent is redacted, not deleted).	Deleting what must be kept is as wrong as keeping what must go.
8 Double-check	A simulator "applies" the plan to a copy and confirms nothing comes back and nothing required is lost. A validator re-reads raw files and checks format, order and dates.	Two independent checks catch our own mistakes before submitting.

5. Where AI (LLMs) is used, and where it isn't

> **Key point:** the run that produces output.csv makes **0 AI calls**. The decisions come from rules plus evidence, so the run is repeatable and explainable. AI helps in three bounded places, and none of them can secretly change an answer.

AI use	Model	What it does	How it helps	Safety
Document classifier (`classify_documents.py`)	OpenAI gpt-5.5 + gpt-5.4-mini	Reads business documents with no explicit status (e.g. a customer statement PDF) and decides which policy category they are (e.g. "USER_EXPORT → delete").	Unblocked a case: the statement PDF was a leftover export that must be deleted.	Saved only when **both models agree**, stored in a file tied to the document's fingerprint (hash), then reused offline.
Reviewer (`llm_review.py`)	OpenAI	A second opinion: reads one case's evidence and our answer, and flags problems.	Finds suspicious cases for humans to check.	Advisory only. Its 3 "major" flags turned out wrong when checked against raw files, which is why we don't let it decide.
Investigator (optional)	OpenAI or Gemini	A read-only agent that can call tools ("read this file") to dig into one hard case.	Explains gaps.	Cannot change accepted answers.

5b. Why we chose each step (and what we rejected)

Decision	Alternatives	Why we chose it
Rules + evidence engine decides; LLM only assists	Send each case to an LLM and ask for the answer	Graders compare exact steps and evidence IDs. An LLM can invent IDs, varies between runs, and costs money on 518 cases. Rules give the same, cited answer every time, so they're cheap, auditable and explainable.
Read every file into "facts" first, once	Search files per case on demand	One index for all 94 files is faster, and nothing is skipped. Each fact keeps its source file, so every step can cite real evidence.
Treat activity log "SUCCESS" as a claim	Trust the log	That is the whole problem: logs said deleted while copies survived (case_16). Only snapshots and readbacks count as proof.
Hard time cutoff (audit_at)	Use all evidence	Required by the rules. Later evidence would "see the future" and give wrong answers for earlier audits.
Resolve old emails/phones via the audit log	Match only the current email	Old systems still store old identifiers; matching only the current one misses ghost copies.
Small policy interpreter that reads deletion_rules.csv	Hard-code one if-statement per rule ID	If the rules file changes, behaviour follows automatically. The rule ID is kept only as a label.
Fix order: stop syncs/restores, then delete, then verify	Delete first	Deleting first lets an active sync or backup restore bring the data straight back.
Two independent checkers before writing output	Trust the planner	A checker that reuses the planner's logic can't catch its mistakes. Separate code re-reads raw files and simulates the plan.
AI document classification needs 2 models to agree, then is cached	One model's answer, called live each run	Agreement reduces errors; the cache makes the run reproducible offline and free.
Learn conventions only from the 32 answer-key samples, tested on held-out halves	Guess rules from intuition, or hand-fix rows	The answer key is the only ground truth. Testing on held-out halves avoids rules that only fit the 32. Hand-fixing rows would be cheating and wouldn't generalise.
Don't guess when evidence is missing (128 rows)	Fill the account in by pattern	Every pattern tested was right only about 33% of the time. A wrong step costs points; a missing check costs less.
Parallel agent team with memory	One long AI chat	Research, coding, testing and review ran at the same time. Memory files stop agents repeating mistakes, and each change was accepted only if the scores improved.

6. How the project was built: AI coding agents

Two AI coding assistants built the code. **Codex** (OpenAI) built the first full pipeline; then **Claude Code** (Anthropic) took over and ran a **team of 5 agents in parallel**. Each has a role, an input, an output file, and a **memory file** so lessons carry over between runs.

Agent	Job	Example contribution
Researcher	Web research; only claims that ≥2 recent sources agree on	Confirmed that a published retention schedule is enough (EDPB 2026, NIST 2025, ICO, CNIL).
Implementer	Changes the engine; accepted only if the samples strictly improve	Added checks for already-compliant records: exact answers 9 → 18.
QA	Built **Harness 2.0**, the scorecard that tests everything	Found that each case's evidence sits inside its own date window.
Reviewer	Read-only code review against the rules	Found that the release step refused to write output.csv at all.
Output	Packaging and the clean re-run test	Proved code.zip re-creates output.csv byte for byte.

Definitions: `.claude/agents/*.md`. Memories: `.claude/agent-memory/`. Both are shipped inside code.zip under `agents/`.

7. The story: what went wrong and how we fixed it

Stage	Undecided cases	What we learned
Codex ends	442 of 518	The code was too strict: it demanded proof the data didn't need, so it refused to answer.
Drop the "every case needs an account record" rule	207	That was our own assumption, not in the rules.
Security logs are kept unless an explicit expiry date has passed	5	All 24 sample answers with security logs keep them; the rule names a "supplied retention period", not a date on every record.
Restorable backup = risk; found legal hold = keep; AI classifies the statement PDF	0	The last blockers, fixed with general rules (never case-by-case hacks).

> **Lesson to tell an interviewer:** "Unresolved" did not mean the data was missing. It meant our code was over-cautious. We fixed each assumption only after the 32 answer-key cases proved it wrong.

Other improvements found by checking against the answer key

- **Check already-compliant records too:** official answers include a check for every relevant record, not only the broken ones.
- **"si" in Spanish can mean "whether":** "check *si* my data was deleted" is a question, not a condition.
- **Catalog links only count inside the case's own date window:** learned from the data's structure and confirmed on samples.
- **A support message showing "[REDACTED]" is proof of redaction** (one more correct check).
- **What we refused to guess:** 128 rows have no account check because nothing in the data links the account. Every rule we tried was right only about 33% of the time (chance), so we left those rows alone.

8. How we tested (and what the numbers mean)

Test	Result	Meaning
32 answer-key samples: status	31/32	We chose COMPLETE / INCOMPLETE / NOT_APPLICABLE correctly 31 times.
32 samples: exact full answer	20/32	Every step identical to the official answer.
Step precision / recall	0.883 / 0.819	88% of our steps are right; we found 82% of the official steps.
False COMPLETE · wrongful deletion	0 · 0	We never said "done" when it wasn't, and never deleted something that must be kept: the two most dangerous errors.
1,024 shuffled variants	992/1024 · 0 changed	The samples re-run with CSV rows shuffled 32 ways. Answers never changed, so the system is deterministic.
30 hand-made edge cases	28/30	Tricky scenarios we wrote ourselves (syncs, restores, legal holds).
Automated tests	225 pass	Unit and regression tests.
Full 518 run	518 rows, 0 errors	Every row passes format, order, date and evidence checks.
Human-style check of 25 answers	20 ok, 1 fixed, 4 unsure	A QA agent compared answers line by line with the raw files.

9. What we submit

Field	File	Contents
code.zip	`upload/code.zip`	Code, tests, prompts, README, evaluation reports, agent files. No API keys, no dataset.
output.csv	`upload/output.csv`	518 answers.
chat_transcript	`upload/log.txt`	Both sessions: 23 Codex entries + 7 Claude Code entries.
AI interview	this guide + `Ghost_Data_Audit_Brief.pdf`	For your preparation.

Your OpenAI key lives only in a local `.env` file, which is never zipped. Graders use their own key via `.env.example`.

10. Ready-made answers

> **30 seconds:** "Dashboards say 'deleted', but copies survive. Our agent reads 94 evidence files, works out who the person is, including old emails, and looks only at what was known at the audit time. It then applies the deletion policy and writes a fix plan. Two independent checkers verify every answer. The final run makes no AI calls; AI only classifies ambiguous documents (two models must agree) and reviews results."

> **Why not just use an LLM for everything?** "Audits must be repeatable and traceable to evidence. Rules plus evidence give the same answer every time, with a citation. We use LLMs where reading is genuinely fuzzy, and we cache the result so it's reproducible."

> **What would you improve?** "128 rows are missing an account check because the data doesn't link the account. With more labeled examples we could learn that link instead of guessing."

11. Glossary (every term, short)

Ghost data
A copy of personal data that survives after a deletion was reported as done.

Case
One deletion request to judge (518 in total).

audit_at / cutoff
The moment we judge a case; evidence created later must be ignored.

reported_status
What the dashboard claims (COMPLETE, PARTIALLY_COMPLETE, NOT_STARTED). Not proof.

Claim vs evidence
A claim is a message saying something happened ("SUCCESS"); evidence is a snapshot or record showing the actual state.

Readback
Evidence that shows a record's actual current state (e.g. the row is in the database snapshot).

Fact
One observation extracted from evidence: system, record, state, time, and source file.

Resource / record
One stored item, e.g. `account_994` in `account_service`.

Replica
A copy of a database kept in sync with the main one; deleting from the main one doesn't always delete the copy.

Search index
A copy of data kept so searches are fast; it can keep showing a deleted person.

Data flow / sync
A background job copying data between systems; if active, it can re-create deleted data.

Backup / restore risk
A saved archive that could put deleted data back; must be blocked from restoring.

Tombstone
A minimal leftover account record keeping only the internal user_id, so other kept records still point somewhere.

Legal hold
A record that must be kept for legal reasons (e.g. a payment dispute).

Retention period
How long a record must be kept (e.g. security logs).

Delete / Redact / Anonymize / Retain
Remove completely / keep but blank out identifiers / keep but make unlinkable to the person / keep as is.

FINDING / REMEDIATION / VERIFICATION
What's wrong / the fix / the check that proves the fix (or proves it's already fine).

evidence_id
The file, activity or audit ID that proves a step. Empty for future fixes and checks.

Catalog link
A `LINK_RESOURCE_IDENTITY` activity row saying "this record belongs to this person".

Request window
The time between when the person asked and the audit time. Each case's own evidence appears in it.

Pipeline
The chain of steps from reading files to writing output.csv.

Ingest / parse
Reading files and turning them into structured facts.

OCR
Optical character recognition: reading text from screenshots.

Policy interpreter
The code that applies deletion_rules.csv to each record.

Simulator
Applies our plan to a copy of the data and checks nothing comes back and nothing required is lost.

Validator
An independent checker for format, order, dates and citations.

LLM
Large language model (e.g. GPT-5.5): AI that reads and writes text.

Two-model agreement
Two different models must give the same answer before we accept it.

Cache / offline replay
Saving the AI's answer to a file so later runs reuse it without calling the AI.

Hash / sha256 / fingerprint
A unique code for a file's exact contents; if the file changes, the code changes.

Agent / subagent
An AI assistant given a role and tools, working on part of the task.

Agent memory
A notes file each agent reads first and updates, so knowledge persists.

Harness 2.0
Our one-command test system: samples, edge cases, and checks on all 518 rows.

Fixture / edge case
A small hand-made test scenario with a known correct answer.

Metamorphic test
Change something that shouldn't matter (e.g. row order) and confirm the answers don't change.

Precision / recall
Share of our steps that are correct / share of the correct steps that we found.

False COMPLETE
Saying "fully deleted" when it isn't; the most dangerous error.

Wrongful deletion
Planning to delete something the policy says must be kept.

Overfitting
A rule that fits the 32 known answers but fails on new cases; we avoided it by testing rules on held-out halves.

Deterministic
The same input always gives the same output.

.env / API key
A private local file holding the secret key used to call OpenAI; never shared or zipped.

log.txt (chat transcript)
The running record of every conversation turn with the coding agents, required for submission.